3D Printing Enclosures

From STEP export to printed box — workflow for hardware engineers

STEP • CAD • FDM • SLA • Tolerances

Audience: Engineering Students & Makers

Tools • Material • Tolerance • Snaps • Heat-set inserts

Why 3D Print Your Own Enclosure?

From PCB to product in days, not weeks

Speed

Design + print in 24 h.
Injection mold = 6+ weeks.

Iteration

Try 5 variants for ₹500
vs ₹5 L of tooling.

Custom fit

Exact fit for your PCB,
battery, connectors.

Low volume

Up to ~1000 units economical.
Above → injection mold.

Prototyping

Demo to investors, customers,
user-test reviewers.

Personal use

One-off projects:
ideal for printed parts.

3D Printing Technologies

Pick the right tech for the part

Tech	Material	Resolution	Cost	Best for
FDM (filament)	PLA, PETG, ABS, TPU	0.1-0.4 mm	Cheapest	Enclosures, brackets
SLA / DLP (resin)	UV-cured resin	25-100 µm	Mid	Detail, smooth surfaces
SLS (powder)	Nylon, PA12	≤ 0.1 mm	High	Functional prototypes
MJF (HP)	PA12 nylon	≤ 0.1 mm	High	Production-quality prototypes
Metal 3D (DMLS)	Stainless, Ti	≤ 50 µm	Very high	Brackets, heat sinks

Student recommendation

Start with a ₹25-50k FDM printer (Bambu A1, Prusa Mini, Anycubic Kobra). PLA filament. ~₹300/kg. Enough for 90% of enclosure prototypes.

PCB → Enclosure Workflow

From KiCad to printed part

- 1. In KiCad: File → Export → STEP File — exports your PCB as 3D CAD
- 2. Import STEP into a CAD tool (Fusion 360, Onshape, FreeCAD, SolidWorks)
- 3. Design the enclosure AROUND the PCB — match mount holes, port cutouts
- 4. Add ribs, heat-set inserts, snap-fits as appropriate
- 5. Export STL or 3MF for slicer
- 6. Slice in PrusaSlicer / Bambu Studio / Cura — generates G-code
- 7. Print, check fit, iterate
- 8. Final design = STEP, send to fab if going to injection mold later

CAD Tools

What to use to design the enclosure

Fusion 360

Free for personal use.
Best commercial tool.

Onshape

Cloud-based, free for
public projects.

FreeCAD

100% open-source.
Learning curve steeper.

SolidWorks

Industry standard.
Paid, professional.

Tinkercad

Browser-based,
simple shapes only.

OpenSCAD

Code-defined CAD.
Great for parametric.

Tolerances & Fit

Design for printer accuracy

Feature	Recommended size	Notes
Min wall thickness	1.2-2 mm	Below = fragile
Min hole diameter	2 mm	FDM drills oversize
Press fit clearance	0.2-0.4 mm	Test and adjust
Loose fit clearance	0.5 mm+	Connector openings
Snap-fit beam	1-3 mm	0.5-1 mm deflection
Bridge length	≤ 30 mm	Else sag in middle
Overhang angle	≤ 45° unsupp	More needs support
Layer height	0.2 mm typical	Lower = smoother, slower

Useful Enclosure Features

Make the print fit the PCB & feel professional

Heat-set inserts

Brass threaded inserts melted in.
Reusable screw holes.

Snap-fits

Lid clips onto bottom
with small flexing tabs.

Living hinges

Single piece + thin hinge —
slim cover.

PCB standoffs

Posts that match
Mounting hole positions.

Cable strain-relief

Channel + clamp
for inbound USB cable.

Light pipes

Translucent feature
over LEDs.

Material Selection

What to print in

Material	Temp limit	Strength	Best for
PLA	55 °C	Brittle	Prototypes, look + fit
PETG	70 °C	Tough	Mechanical parts, food-safe
ABS	100 °C	Strong	Outdoor, heat exposure
TPU	70 °C	Flexible	Gaskets, bumpers
Nylon	100 °C	Very strong	Gears, brackets
ASA	100 °C	UV-stable ABS	Outdoor enclosures
PC	115 °C	Very tough	Engineering applications
Resin (SLA)	60-80 °C	Brittle	Detail, smooth surfaces

Common Mistakes

What sends prints back to the printer

- Forgetting to update CAD when PCB grew by 2 mm
- Designing exact-fit holes — print shrinks 1-2%, parts won't fit
- Sharp corners that look great in CAD but warp in PLA
- Connector openings too tight — micro-USB / USB-C pin counts vary
- No air vents → moisture builds, condensation
- Heat-set inserts too small/large for the brass insert chosen
- Snap-fit too rigid — breaks first close
- Overlooking weight & shipping cost

Outsourcing 3D Printing

When you don't own a printer

Shapeways

Online 3D printing service.
Many materials.

PCBWay

3D printing + PCB combo orders.
FDM + SLA.

JLC3DP

Cheap Chinese 3D printing,
good for prototypes.

Local makerspace

Bengaluru, Pune,
Chennai, Delhi have makerspaces.

3D Hubs / HUBS

Crowdsourced —
local printer near you.

Print-on-Amazon

Slow but accessible.
Good for one-off.

Heat-Set Inserts & Threaded Holes

Professional-looking screw mounts

- Brass insert pressed into heated hole via soldering iron
- Threaded by manufacturer — M2.5, M3, M4 standard
- Provides strong, reusable thread that won't strip with plastic
- Hole diameter slightly smaller than insert OD (~0.2 mm)
- Heat-set tool ($20-50): controlled temperature soldering iron tip
- Indian suppliers: TR-Fastenings, Robu, Amazon
- McMaster-Carr / RuthEx (Germany) for premium inserts
- Pre-design: leave room for the boss around the insert

Integration with PCB Workflow

Keep CAD and PCB synchronised

- Use single source of truth — STEP from KiCad imported into Fusion 360
- Lock the PCB outline + mount holes BEFORE designing enclosure
- Tag the assembly: 'PCB v1.0 + Enclosure v1.0' linked
- Test virtually: shrinkwrap the enclosure around PCB in CAD
- Print the entire assembly: PCB + enclosure together to verify fit
- Stencil paste application: assemble enclosure → reflow PCB → close
- Version-control the STEP / F3D file alongside the KiCad project
- On rev change: re-export STEP, update CAD, re-print, re-test

IP Rating — Sealed Enclosures

Dust and water resistance

- IP code: IPxy where x = dust (0-6), y = water (0-9K)
- IP54 — splash-proof, indoor outdoor
- IP65 — fully dust-proof, water jets
- IP67 — submersible 1 m for 30 min
- IP68 — continuous submersion
- 3D printed enclosures rarely achieve > IP54 without help
- Gaskets (TPU printed or silicone) — bring up to IP65
- Conformal coating + sealed enclosure → IP67 possible

Key Takeaways

Key takeaways from this lecture

STEP from KiCad

Export every revision.

Design CAD around PCB

PCB is source of truth.

Test tolerances

Print parts at scale, measure.

Heat-set inserts

Reusable threaded holes.

FDM PLA for protos

Cheapest, easiest start.

Iterate fast

5 prints in 5 days beats 1 mold in 5 weeks.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- Fusion 360 free educational license
- PrusaSlicer / Bambu Studio (free)
- Andrew Maker / CNC Kitchen YouTube
- Hackaday — 3D printing for hardware
- KiCad STEP export documentation
- Local makerspaces: WorkBench Projects, Maker's Asylum, IIT Madras
- Hub3D / Shapeways / JLC3DP services
- McMaster-Carr / RuthEx heat-set inserts

Thank you • Build something this week.